

Philip-Chandan Imaging Pipeline

TCGA-BRCA longitudinal MRI: raw export, napari tumor ROI, PDE handoff

QBIHack breast-cancer-sim | Generated 2026-06-28

Philip-Chandan owns the radiomics pipeline: pull real breast MRI from TCIA, validate and stack DICOM into 3D numpy arrays, define tumor ROIs with a napari aligned-bbox workflow, and hand raw volumes plus expert masks to Vinesh for PDE simulation. Resample, normalize, and crop are Vinesh's scope (handoff_contract.json v1.1.0, Option B).

1. Patient selection criteria

We needed real TCGA-BRCA cases for a Luminal A vs Basal-like demo with longitudinal MRI, not a single timepoint. Each candidate had to pass several gates:

Criterion	Why it matters
MRI on TCIA	Many TCGA patients have genomics but no public imaging
Longitudinal MR (>=2 study dates)	Baseline + follow-up for growth comparison
PAM50 subtype match	Luminal A and Basal-like labels align with cBioPortal
Genomics for Praneeth	ER/PR status and survival on GDC/cBioPortal
Usable contrast series	Post-contrast T1 (e.g. VIBRANT) for tumor visibility

Our first picks failed: TCGA-BH-A0BR and TCGA-A2-A04P had no MRI on TCIA. Only 19 of 139 TCGA-BRCA patients with MRI on TCIA have multiple MR studies. Patient selection was a real constraint.

Rev2 primary pair (cohort/cohort.json)

Subtype	TCGA ID	Baseline	Follow-up
Luminal A	TCGA-AR-A1AX	2002-09-12	2003-09-24 (~12 mo)
Basal-like	TCGA-AR-A1AQ	2001-11-21	2003-05-07 (~17 mo)

2. Finding patients - cohort_discovery.py

Before locking IDs in cohort.json, we built cohort/cohort_discovery.py to automate discovery and validation against public APIs:

- TCIA NBIA REST API - list MR series, group by study date, flag longitudinal cases, prefer contrast series
- cBioPortal - PAM50 subtype, ER/PR, overall survival (aligned with Praneeth's genomics fields)

CLI workflows

```
python ../cohort/cohort_discovery.py audit
python ../cohort/cohort_discovery.py find-longitudinal --subtype "Luminal A"
python ../cohort/cohort_discovery.py recommend-pair
```

Each patient gets a structured report: imaging availability, study span, PAM50 match, missing genomics fields, and a pass/fail ok flag. This tool justified the rev2 pivot and gave Praneeth the same TCGA barcodes for GDC queries.

3. Download - download_tcia.py

Once patients are locked, download_tcia.py pulls DICOM into a consistent layout:

```
data/raw/tcia/
  luminal_a/TCGA-AR-A1AX/
    2002-09-12/ # baseline
    2003-09-24/ # follow-up
  basal/TCGA-AR-A1AQ/
    2001-11-21/
```

2003-05-07/

For each study date the downloader:

- Lists MR series in collection TCGA-BRCA
- Picks the best series - prefers post-contrast (+C, VIBRANT, T1); deprioritizes calibration/localizer scans
- Downloads via idc-index (primary) with tcia-utils NBIA fallback

```
python ../download_tcia.py --all-primary --longitudinal
```

4. Validate - validate_series() in tcia_extractor.py

Download alone is not enough. Before export, validate_series(dicom_dir) checks the local series:

Check	What we catch
DICOM slices present	Empty or non-image folders
Consistent Rows/Columns	Mixed slice dimensions
Unique InstanceNumber	Duplicate or corrupted ordering
SimpleITK read succeeds	Metadata OK but ITK cannot stack
Slice count, shape, spacing_mm	Recorded in validation report for QC and sidecar JSON

extract_volume() refuses to run if validation fails. Export scripts call validate_series first and abort with explicit errors.

Visual QC

qc_slice_plot.py writes middle-slice PNGs per volume to data/qc/slice-plots-philip-chandan/, including optional overlay PNGs with a lime contour over high-intensity voxels (QC visualization only, not tumor segmentation).

Figure 1 - Spike validation slice

luminal_a_TCGA-AR-A1AX_baseline z=27 (>90th pct contour)

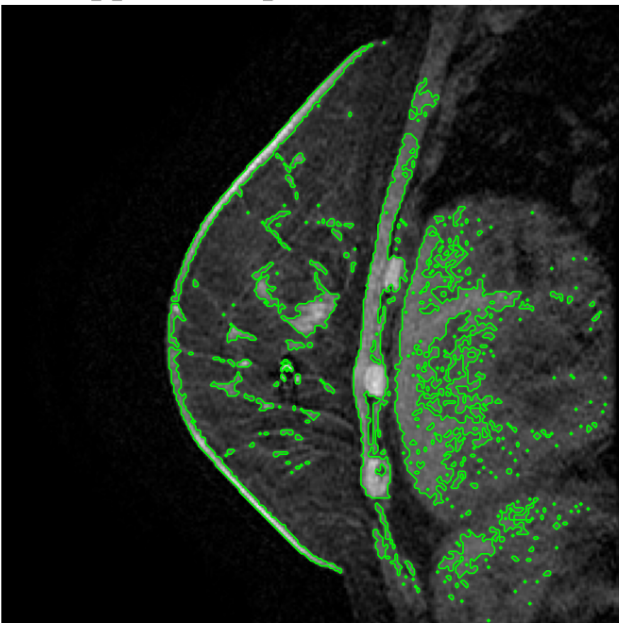


Figure 1. Raw MR slice for TCGA-AR-A1AX baseline (2002-09-12) after DICOM validation and 3D stacking. Lime contour marks voxels above the 90th percentile (QC only).

5. Export - raw volumes for Vinesh

After validation passes, we export raw 3D arrays, not PDE-ready volumes. Resample, normalize, and segmentation are Vinesh's scope per handoff_contract.json.

- Single case: export_raw_extract.py
- Batch (all primaries x timepoints): export_all_raw.py

For each patient x timepoint slug (e.g. luminal_a_TCGA-AR-A1AX_baseline):

- Resolve DICOM path from cohort.json
- Run validate_series
- extract_volume_with_spacing() - SimpleITK DICOM to float32 numpy, shape (Z, Y, X), native spacing preserved
- Write .npy, JSON sidecar, and QC plot

Output	Path
Raw volume	data/processed/raw-extract-philip-chandan/{slug}.npy
Metadata sidecar	.../{slug}.json (shape, spacing_mm, contract_version)
QC plot	data/qc/slice-plots-philip-chandan/{slug}_mid-z[-overlay].png
Manifest	manifest.json v1.1.0 - subtype, timepoint, slug, paths

Handoff contract - what we export vs what Vinesh prepares

Philip-Chandan (raw extract + ROI)	Vinesh (PDE input)
Raw MR intensities, not normalized	Resample/crop to max 64^3, 1 mm spacing
Spacing in JSON sidecar	Min-max normalize to [0, 1]
Axis order (Z, Y, X)	Apply napari expert mask; keep continuous density inside ROI
Napari aligned-bbox mask -> segmentations/	Zero background; center crop on tumor COM

Figure 2 - Baseline subtype comparison

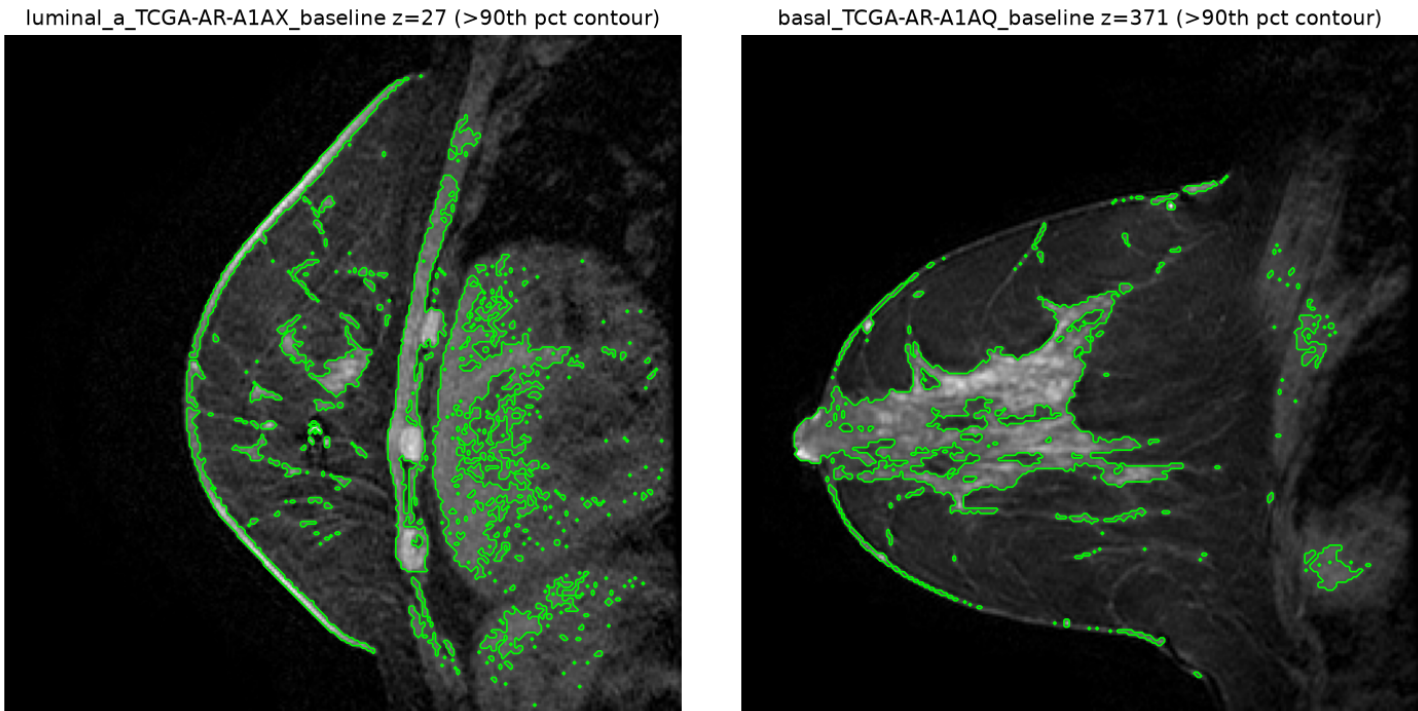
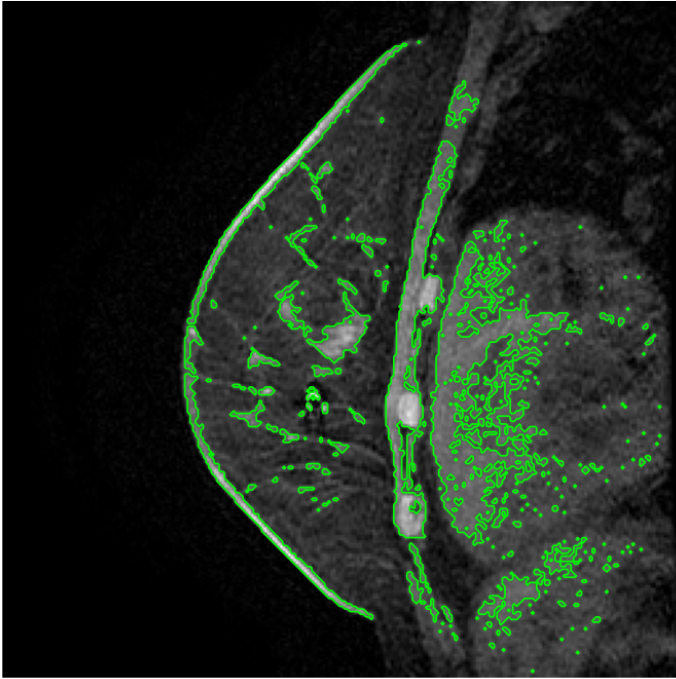


Figure 2. Baseline comparison: Luminal A (TCGA-AR-A1AX) vs Basal-like (TCGA-AR-A1AQ). Lime contour highlights enhancing tissue (90th percentile, QC only).

Figure 3 - Longitudinal Luminal A

luminal_a_TCGA-AR-A1AX_baseline z=27 (>90th pct contour)



luminal_a_TCGA-AR-A1AX_followup z=40 (>90th pct contour)



Figure 3. Longitudinal Luminal A: baseline (2002-09-12) vs follow-up (2003-09-24). Contour overlay on the slice with the most enhancing voxels per volume.

6. Tumor ROI - TCIA center annotation and napari aligned-bbox workflow

TCGA-Breast-Radiogenomics publishes expert lesion files (*.les) for baseline DCE studies. These are not hand-painted 3D contours: radiologists marked an approximate tumor center on ClearCanvas, and the UChicago workstation ran fuzzy c-means (FCM) auto-segmentation inside a small bounding cuboid. For our rev2 primaries the cuboid is only ~31-34% filled (~1.3k-2.7k FCM voxels); in napari the annotation reads as sparse dots at phase-1 z, while later DCE phases show brighter enhancement in the same region.

The useful ground truth is therefore the expert center plus bounding box (Y/X/Z bounds), not the sparse FCM voxels alone. Global Otsu on the full breast volume fails badly against these masks (Section 8). Instead we use an aligned-bbox napari workflow to manually threshold inside the cuboid and export a singly-connected tumor ROI for simulation.

Aligned-bbox workflow (validation/view_aligned_cuboid_napari.py)

- Split stacked VIBRANT into DCE phases; align P2-P3 z-band slabs rigidly to phase 1 (late DCE tail skipped)
- Inside the tight .les Y/X bounding box on aligned P2-P3 slabs: sweep intensity threshold and track the center-connected bright region from the expert center
- In napari: adjust threshold slider, optionally allow a small connectivity gap, jump to elbow on the connected-fraction curve, visually confirm tumor on subtraction phases
- Export mask -> local .npy; publish_expert_mask.py writes brain-parity segmentations/{slug}_mask.nii.gz and links segmentation_path on the raw JSON sidecar

Rev2 baseline	Phase @ threshold	Mask voxels	Notes
TCGA-AR-A1AX (Luminal A)	P2 @ 0.35	~1,798	Center-connected napari export
TCGA-AR-A1AQ (Basal-like)	P2 @ 0.412	~4,683	Rim lesion; +387 voxels necrotic-core fill

Figure 4 - Threshold curve inside expert bounding box

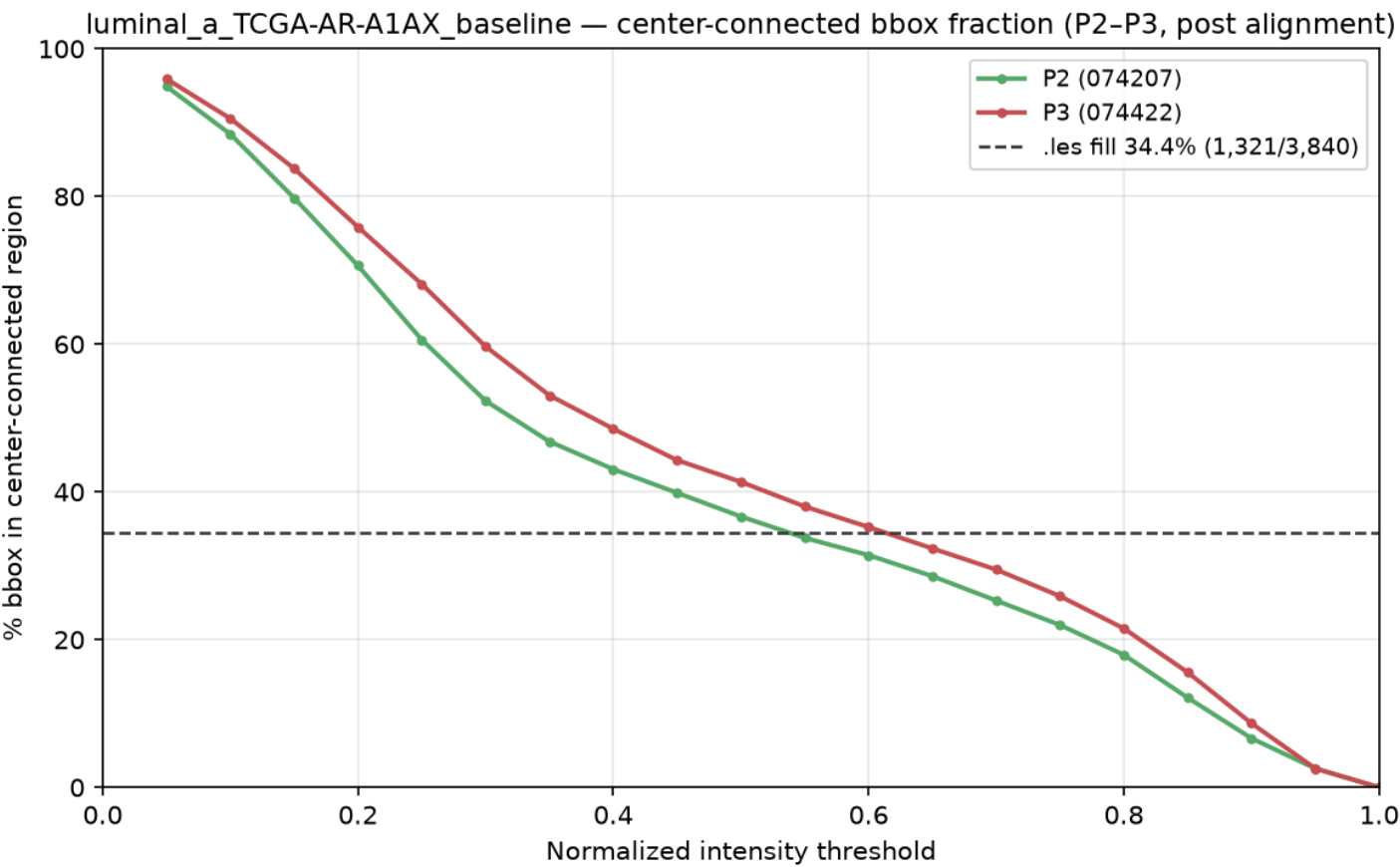
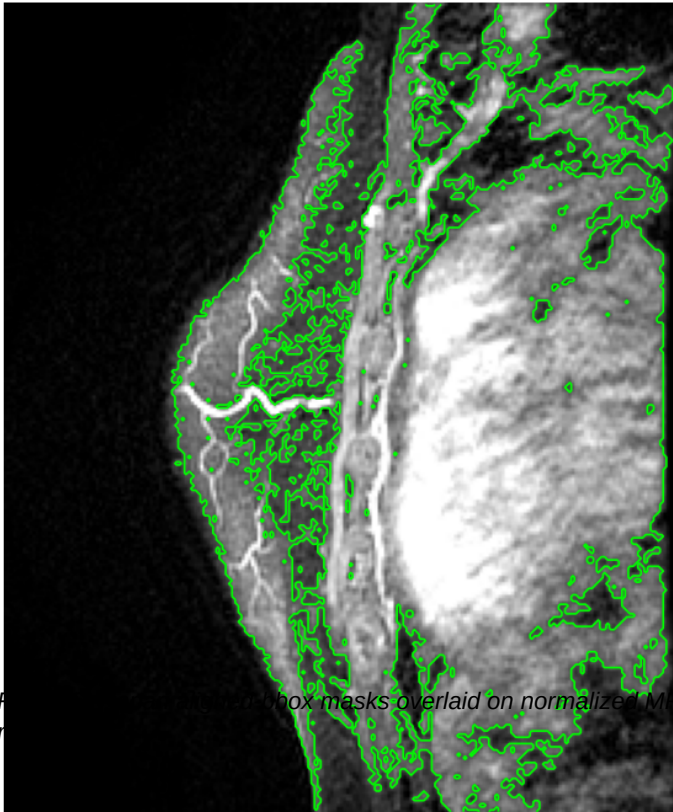


Figure 4. TCGA-AR-A1AX baseline: center-connected bright fraction vs threshold inside the aligned .les bounding box (P2-P3). Elbow guides manual threshold choice in napari.

Figure 5 - Exported tumor mask on MR

TCGA-AR-A1AX baseline z=224 (mask vox=3500566)



TCGA-AR-A1AQ baseline z=35 (mask vox=1698339)

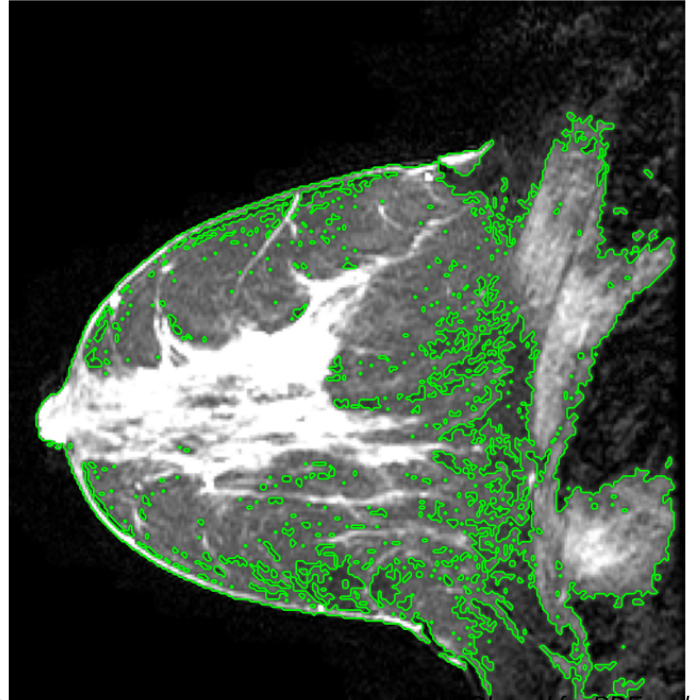


Figure 6 - Expert mask on normalized resampled volume. The green vox masks overlaid on normalized MR (baseline primaries). Lime contour = exported ROI used for PDE and QC figures.

7. PDE simulation input - prepare_pde_input.py (Vinesh)

After Philip-Chandan publishes the napari expert mask, Vinesh prepares PDE-ready volumes in vinesh/prepare_pde_input.py per handoff_contract.json v1.1.0:

- Load raw MR + expert mask from segmentations/{slug}_mask.nii.gz
- Resample both to isotropic 1 mm spacing (scipy.ndimage.zoom)
- Min-max normalize MR intensities to [0, 1]
- Keep continuous normalized density inside the expert mask; zero background
- Crop/pad to max 64^3 centered on tumor center of mass

Continuous tumor values (not a binary mask) are required so the PDE logistic growth term $\rho \cdot u \cdot (1-u)$ can evolve. qc_otsu_plot.py documents the same pipeline for QC figures.

Figure 6 - Expert mask on normalized resampled volume

TCGA-AR-A1AX expert mask z=59 (mask vox=4029)

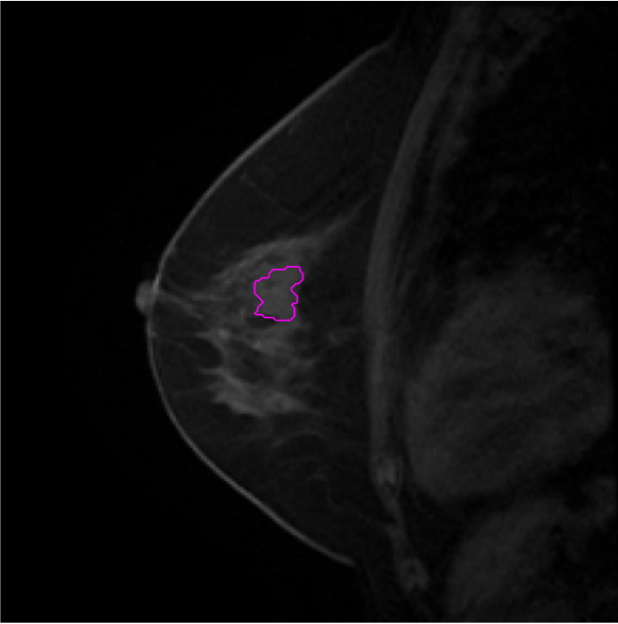
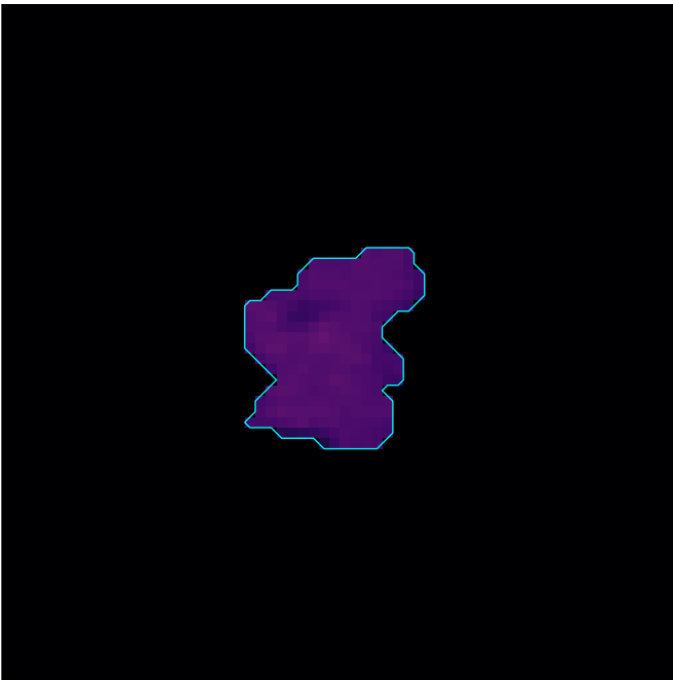


Figure 6. TCGA-AR-A1AX baseline after resample + normalize. Magenta contour is the published napari expert mask (voxels inside ROI); background will be zeroed before crop.

Figure 7 - PDE input after mask and crop

TCGA-AR-A1AX PDE input z=26 shape=[64, 64, 64]



TCGA-AR-A1AQ PDE input z=29 shape=[64, 64, 64]

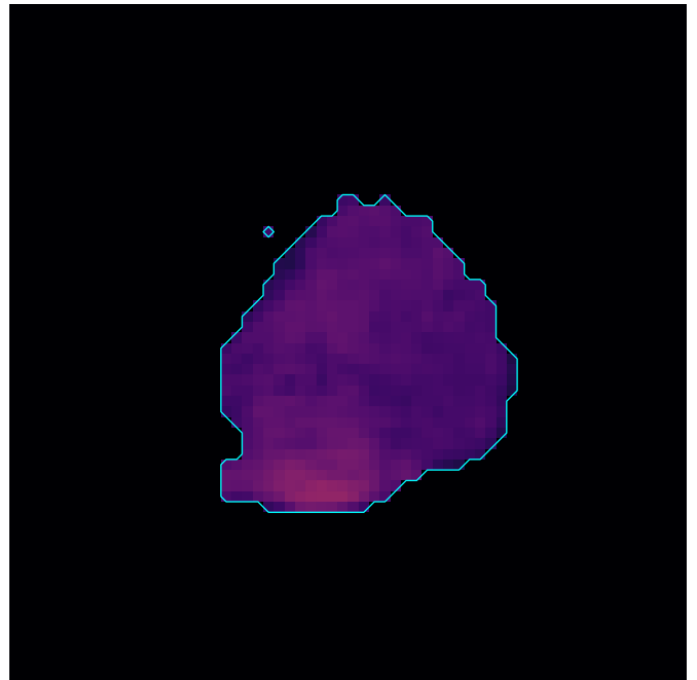


Figure 7. Baseline PDE inputs (64^3 , 1 mm) for Luminal A vs Basal-like. Inferno colormap shows continuous initial tumor burden inside the expert ROI; cyan contour marks tumor voxels > 0 .

8. Why global Otsu was retired - validation vs TCIA radiologist masks

TCGA-Breast-Radiogenomics publishes expert lesion segmentations (*.les) for 91 patients. We downloaded the public ZIP (~103 KB), parsed cuboid masks, and compared an early global-Otsu + largest-component heuristic to the matching DCE series (S2 = VIBRANT for both rev2 baselines). That approach is no longer used for PDE input; it motivates the napari aligned-bbox workflow in Section 6. Follow-up timepoints have no .les annotations.

Slug	Dice	Area (Otsu/.les)	Expert vol	Otsu vol
basal TCGA-AR-A1AQ baseline	0.000	619.4x	6,075	3,762,799
luminal a TCGA-AR-A1AX baseline	0.000	2649.9x	2,927	7,756,222

Dice is near zero on both primaries: global Otsu selects a much larger bright region than the radiologist ROI, often on different z slices than the expert cuboid. Area fraction (Otsu voxels / .les voxels) is ~2650x for Luminal A and ~620x for Basal-like. The napari workflow constrains thresholding to the expert bounding box and center-connected region instead.

Figure 8 - Radiologist .les vs global Otsu (baseline, historical)

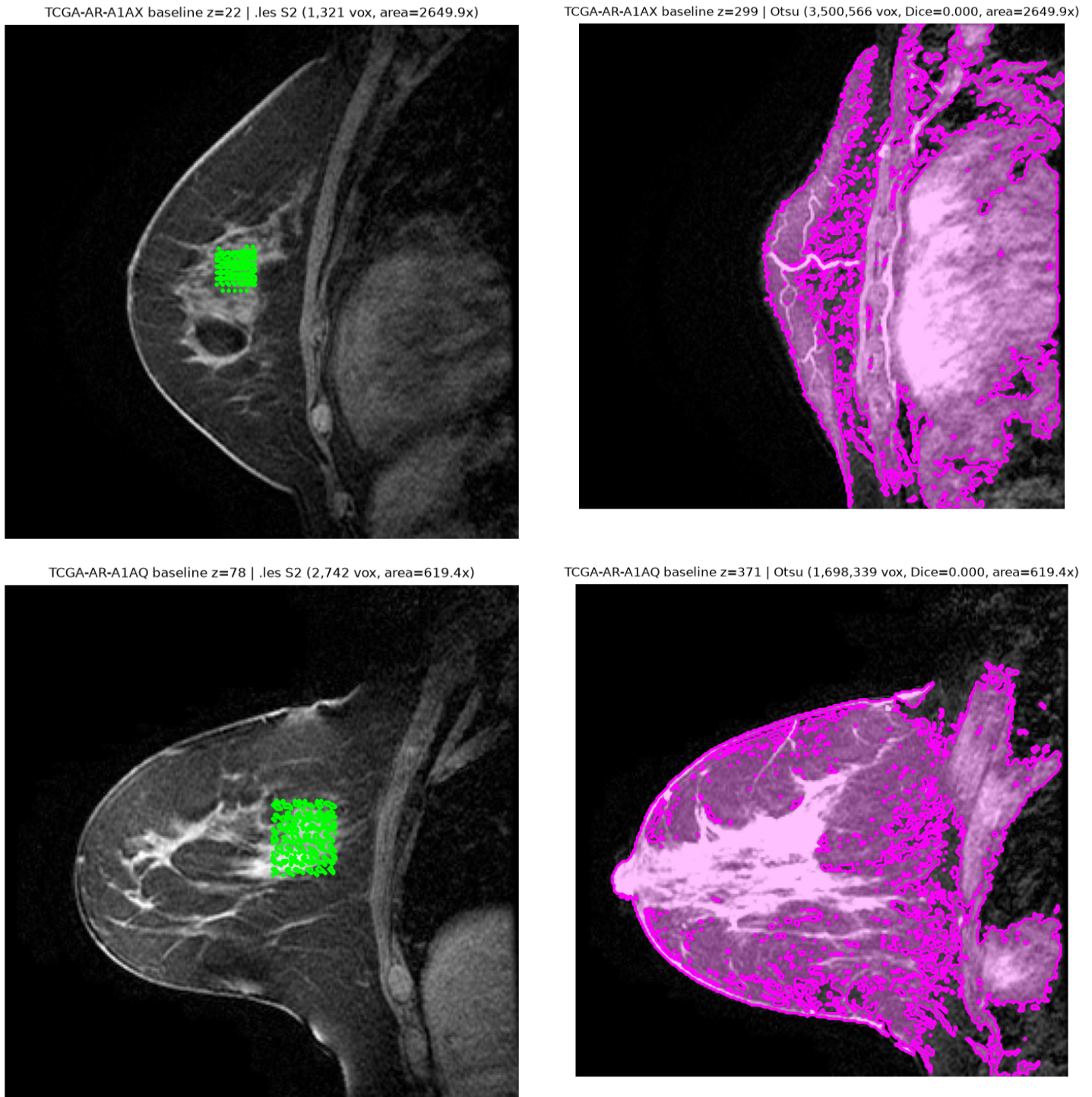


Figure 8. Baseline validation (2x2): top row = Luminal A .les then global Otsu; bottom row = Basal-like .les then Otsu. Green = TCIA .les; magenta = retired Otsu. Each panel uses its own best z slice (masks do not overlap in slice space). Titles show voxel counts and area=Otsu/.les.

9. End-to-end summary

We defined strict patient criteria (longitudinal MRI on TCIA, PAM50-aligned subtypes, genomics for the team), built cohort_discovery.py to search and audit candidates against TCIA and cBioPortal, locked rev2 primaries after rev1 failed imaging availability, downloaded four DICOM timepoints with contrast-aware series selection, validated each series structurally before extraction and visually via slice QC, exported four raw .npy volumes plus JSON sidecars and a manifest, defined tumor ROIs with a napari aligned-bbox workflow inside TCIA expert bounding boxes (center annotation + manual threshold), published expert masks for Vinesh PDE prep into 64^3 simulation inputs, and documented why global Otsu was retired after benchmarking against TCIA radiologist .les masks on baseline VIBRANT

(Dice near zero).

Deliverables

Stage	Deliverable	Count
Discovery	cohort.json rev2 + COHORT.md	2 primaries + backups
Download	DICOM under data/raw/tcia/	4 study folders
Validate	validate_series pass + QC PNGs	4 volumes
Export	Raw .npy + .json + manifest.json	4 slugs
Tumor ROI	Napari aligned-bbox mask + segmentations/	2 baselines
PDE prep	Expert-mask QC + pde-input-vinesh/	4 slugs
Seg. validation	.les vs Otsu benchmark + validation_metrics.csv	2 baselines

Exported volume shapes (rev2)

Slug	Shape (Z,Y,X)	Spacing (mm)
luminal_a_TCGA-AR-A1AX_baseline	[352, 256, 256]	[3.0, 0.8594, 0.8594]
basal_TCGA-AR-A1AQ_baseline	[464, 256, 256]	[3.0, 0.859375, 0.859375]
luminal_a_TCGA-AR-A1AX_followup	[552, 512, 512]	[2.2, 0.5273, 0.5273]
basal_TCGA-AR-A1AQ_followup	[448, 256, 256]	[3.0, 0.9375, 0.9375]